KEDGE dart v34: engineering report

v34 is the self-righting layout from the righting study (kedge/sim_righting): drill pods at the tail folding forward, a low bridle to eye lugs on the pods, a top righting leg instead of the kick leg and foot, and a low centre of mass. All numbers below come from the exact print geometry and the pipeline (source/run_all.sh).

1. Summary

Item	Result
Body	Ø42 x 152 mm; folded envelope x -0.0 to 152.0, y ±40.5, z -21.0 to 21.3 (nothing behind the tail, nothing above the hull but a 0.3 mm knuckle blister)
Mass, centre of mass	248.1 g with catalogue bought masses and 14.6 g nose putty; CG folded [83.15, -0.1, 0.44], anchored [85.0, -0.1, -3.81], leg out [81.15, -0.1, 0.34] (mm; z from the hull axis)
Collision sweep	CLEAN: righting leg 0 to 178.0° in 179 steps, pods 0 to 46.75° in 48 steps, leg x pod cross sweep, every sub-part, exact meshes
Hard stops (mm³ of overlap just past each end)	deploy_at_angle 0.0, deploy_horn_+2.5deg 5.23, fold_at_0 0.0, fold_lug_-3deg 5.57, kick_rest_-0.5deg 0.0, kick_rest_+2deg 58.54, leg_fold_at_0 0.0, leg_fold_-2deg 433.47, ok True
Drills	deploy 46.75° (solved: the motor end in the hub stays 0.5 above flat ground), 43.5 mm deep, entering the soil about 106 mm from the nose tip (near the tail)
Launcher fit	FITS over 1684 launcher states (base/cad/parts/launch.py, parked), the stowed bridle included (section 5)
Wire room	left corridor worst fill [0.27, 122.0], right [0.1, 120.0] (bundle area over free area; under 0.6 is fine)
Printed parts	79.97 cm³, every printed part watertight and one body

2. Layout and why

• Pods: pivot x 132, z 9.5, drill axis |y| 31.15. The study put the pivot at x 140, z 10; x 132 leaves room behind the deploy stop for the leg's pad in front of the flat tail face the pusher needs, and z 9.5 keeps the centre of mass near the axis while the drills still reach 43.5 mm.
• One deploy worm N20 on the axle (motor pointing down), the v33 PPA-CF couplers: they run in the pivot-block bores (the hull carries the pods) and are PRESS FITS with CA on the worm's D stubs (socket Ø3.05, flat at 1.03) and in the arm heads (D bore 7.05): the axle is stiff, so both pods load the one worm together (one pod on its stop cancels the other's soil moment). The pods stop on faces carved in the pivot blocks about 1° past deployed and past folded.
• Bridle eye: a lug on each arm head's outboard 5 mm, 3.9 forward of and 6.8 below the pivot when folded. Deployed it sits behind and below the pivot, about 23 mm above the soil, so the tow pulls the pod onto its deploy stop (moment about half the study's eye; the eye could not hang lower: the launcher's stop springs pass 0.6 mm under it at the shot). No latch (Harry): the worm is self-locking and the study found 0.05 to 0.25 N m on it.
• Drills: the v33 tapered gimlets unchanged, mirrored to point forward, hands kept (left pod right-hand, firmware DRILL_SIGN +1). The auger hub runs on the arm's bushing and its base face on the arm shoulder; pull along the auger goes into the motor's threaded shaft through nut A (this motor is sold for lead-screw use).
• Righting leg: hinge x 33, z 14.5 on the centreline; swing 0 to 178° with a bed 1.5° further on in the nose top. The leg N20 sits under a 0.6 mm skin over its worm box (the knuckle), motor pointing down. The leg's fork has a round cheek on each side of the knuckle; each cheek is keyed on a Ø6 PPA-CF coupler that runs in a bore in the knuckle wall and in the blind bore of a flush skin plug, and takes the trimmed worm stub in a press-fit socket (CA); the cheeks are press fits on the couplers (D bore 6.05). The blade (I-channel, its top IS the hull crown) lies flush in the dorsal groove; the pad is a 2 mm patch of the hull skin on the tail top. Pad centre 113 mm from the hinge (study 128) and about 37 mm wide (study 56): the pad must stay in front of the pusher face and between the folded pods.
• Low centre of mass: the cells in line on the belly (z -10), both worm motors pointing down, DRVs standing low beside the deploy motor, boards on decks right over the cells. The ballast putty is in the nose (flight); righting works from any attitude.
• Sealed body: clamshell halves with the tray's deck 1 edges in a seam groove behind the split line; the nose spigot with its lock rib; couplers in tight bores; the drill leads through tight holes (silicone); the USB-C window tight round the receptacle; the JST headers fill their flank seats; the stub slots under the knuckle are closed by the tray's tongues.

3. Load paths

• Anchored tow: line -> bridle -> eye lugs on the arm heads -> arm -> coupler -> pivot-block bores (hull). The eye moment pushes the pods onto the carved deploy stops; any rest of the soil moment that would fold a pod back is held by the self-locking worm (study: up to 0.25 N m without the eye help).
• Drill loads: side load on the auger hub -> arm bushing; push -> arm shoulder; pull -> nut A on the motor's threaded shaft.
• Righting: leg -> two couplers -> knuckle-wall bores and skin plugs (hull); the worm stubs only turn the couplers. At 178° the blade root lies in its bed in the nose top; folded, the blade rests on the groove floor.
• Launch: pusher -> flat tail end walls of the two halves -> hull -> radial screws -> tray -> cells on the rear plate. Nothing protrudes behind the tail.

4. Mass properties (catalogue bought masses; weigh and update)

Pose	Mass g	CG x, y, z mm
folded	248.1	[83.15, -0.1, 0.44]
deployed	248.1	[85.0, -0.1, -3.81]
leg_out	248.1	[81.15, -0.1, 0.34]

Printed 99.3 g at 4 walls / 40% infill; bought parts and extras 148.8 g (incl. 14.6 g nose putty). Per-part volume, centroid and inertia: mass_props.json.

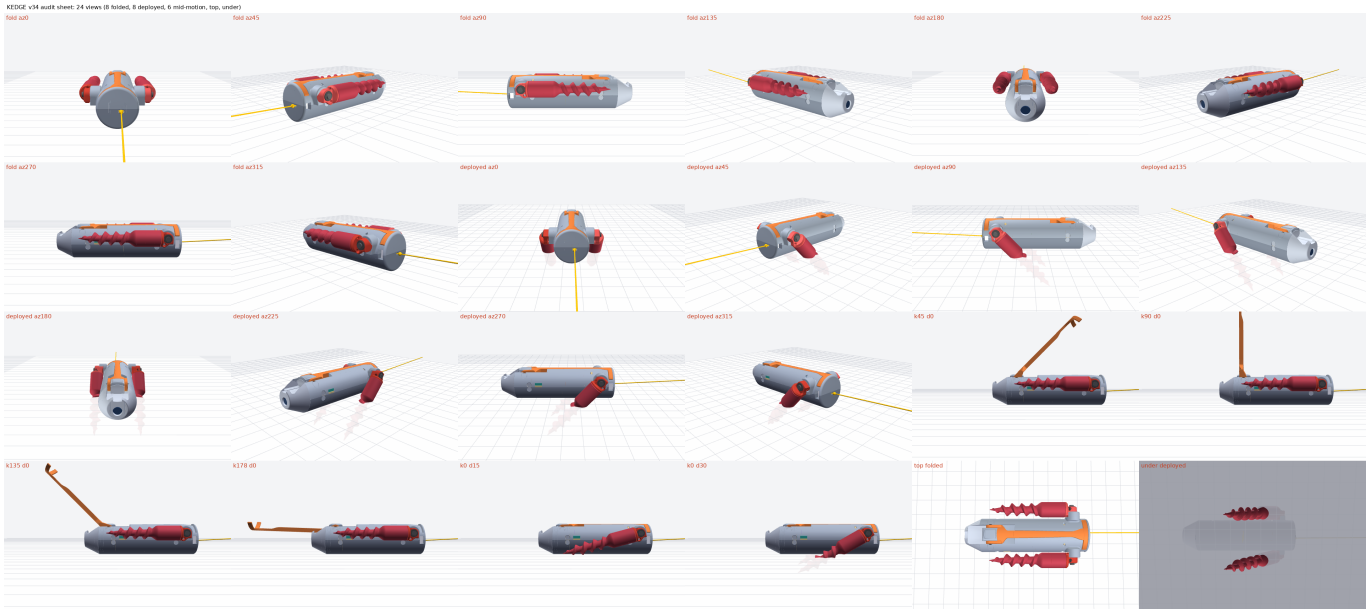
5. Launcher (parked base) and the bridle

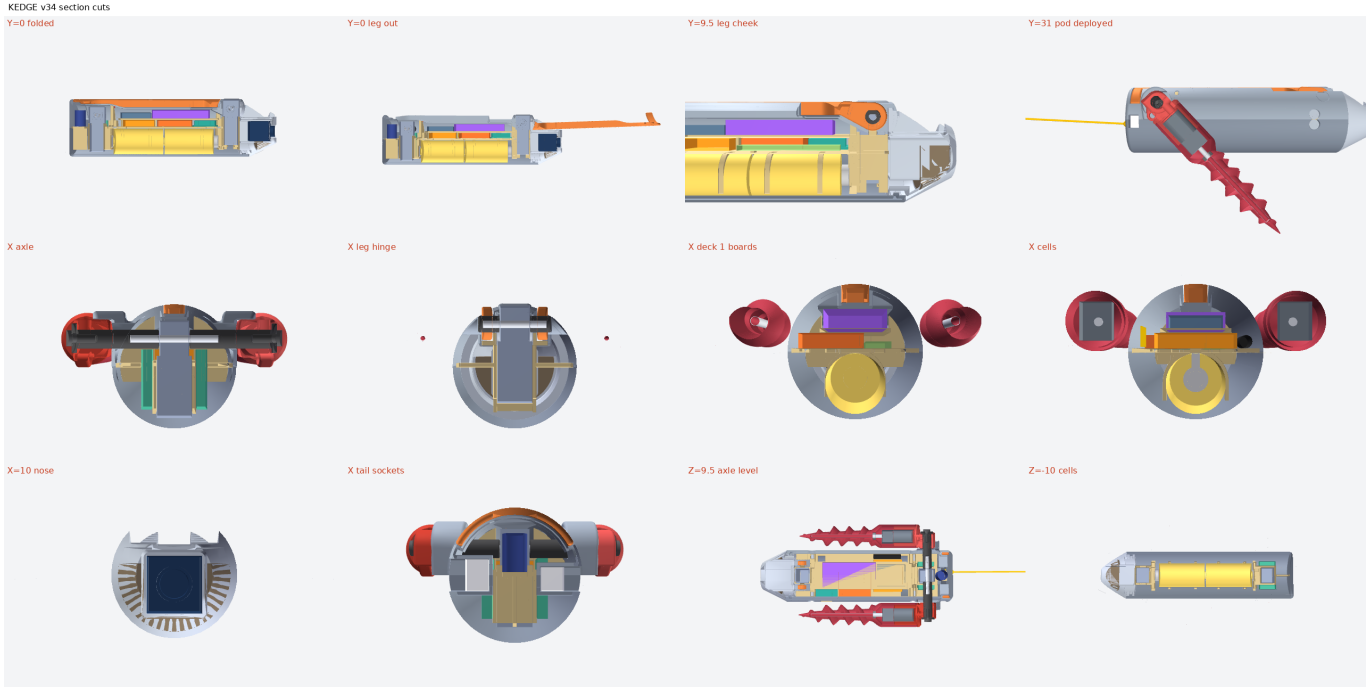
• Fit: FITS over 1684 states. The rest check at home: dart on the cradle {'dart_vs_carriage_mm3': 0.0, 'dart_down_0p5_vs_carriage_mm3': 282.253}
• Bridle: one piece of 1 mm Dyneema, legs 78 mm from the eye lugs to a sliding figure-8 loop in the rover line (both legs share the pull). Stowed on the launcher each leg runs from its eye past the outside of the arm head, climbs to the tail edge at 65 deg from the top, rounds the edge in a groove (1.6 wide, 1.4 deep) and crosses the end wall in a groove to (|y| 4.2, z 3.2), inside the pusher's Ø14 hole; the loop sits on the axis about 26 mm behind the tail face. launcher_fit models the stowed rope (Ø1.2 swept polyline, bowlines and loop) moving with the dart over every base state: rope checks {"tie": {"holes": [[152.0, 1.6, -2.4], [152.0, 5.4, -2.4]], "knot": [154.2, 3.5, -2.4], "hole_d": 2.0, "line_d": 1.2, "knot_r": 2.0}, "ok": true}. Towed, the legs lift out of the grooves and run straight to the loop about 52 mm behind the tail.
• What base needs: nothing for the bridle (the slots listed earlier are no longer needed). Base's line model (launch_line_*) should start at the loop on the axis behind the pusher face; the pusher still pushes the flat tail face within |y| 11 beside the grooves, and the seat pushrod still lands on solid face at (y -15, z +1), more than 2 mm from the right groove.

6. For dart-physics and dart-fw

• dart-physics: flight with the nose putty and the sliding bridle loop, landing, holding (pods at the tail, drills under the tether) and the righting sim with this leg (113 mm, pad 37 mm, hinge x 33, z 14.5) and this mass; mass_props.json has the leg_out pose.
• dart-fw: the kick channel (DRV#1 B, D2/D3) drives the righting leg (0 to 178°, far stop = its bed in the nose); the foot channel (DRV#2 B, D6/D7) is unused; the deploy end stop is the carved stop at the solved deploy angle; the wire list has 45 wires (no foot, no series link).

7. Pictures


KEDGE v34 audit sheet: 24 views (8 folded, 8 deployed, 6 mid-motion, top, under)
fold az0
fold az45
fold az90
fold az135
fold az180
fold az225
fold az270
fold az315
deployed az0
deployed az45
deployed az90
deployed az135
deployed az180
deployed az225
deployed az270
deployed az315
k45 d0
k90 d0
k135 d0
k178 d0
k0 d15
k0 d30
top folded
under deployed


KEDGE v34 section cuts
Y=0 folded
Y=0 leg out
Y=9.5 leg cheek
Y=31 pod deployed
X axle
X leg hinge
X deck 1 boards
X cells
X=10 nose
X tail sockets
Z=9.5 axle level
Z=-10 cells